

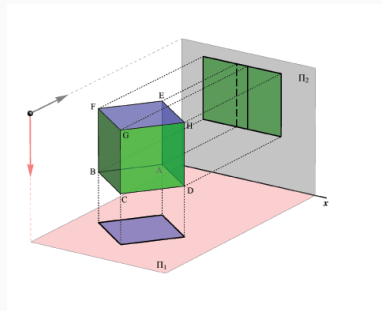
Galerkin and FE approximation

Linear elastodynamics

ME473 Dynamic finite element analysis of structures

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Where do we stand?

Week	Module	Lecture topic	Mini-projects
1	Linear elastodynamics	Strong and weak forms	
2		Galerkin and FE methods	

Summary

- Recap week 1
- Further evidence in favor of the weak form
- Discretisation of the weak form of elastodynamics via Galerkin method
 - Example: longitudinal vibration of a bar
 - Matlab implementation of Galerkin approximation
- Finite element method in global coordinate system

Recommended readings

- ① Gmür, Dynamique des structures (§2.3, §2.4, §3.1 and §3.2)
- ② Neto et al., Engineering Computation of Structures (§2.1, §2.2 and §2.3)

Recap week 1

Strong and weak forms of elastodynamics

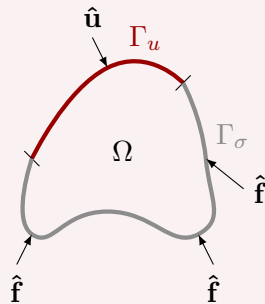
Statement of the linear elastodynamics problem

■ Object:

A solid $\Omega \subset \mathbb{R}^3$ (beam, shaft, plate etc...) with known material properties: $\mathbf{C}$ and ρ .

■ Main features:

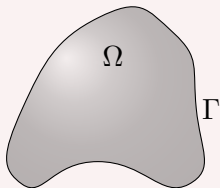
- Acting loads on the body: $\mathbf{f}$.
- Boundary $\Gamma = \Gamma_u \cup \Gamma_\sigma$ (the surface enclosing the solid).
- Boundary conditions:
 - ▶ Prescribed displacements $\hat{\mathbf{u}}$ on the boundary Γ_u
 - ▶ prescribed loads $\hat{\mathbf{f}}$ on the boundary on Γ_σ .
- Initial displacement $\mathbf{u}_0$ and initial velocity $\mathbf{v}_0$ at $t = 0$.



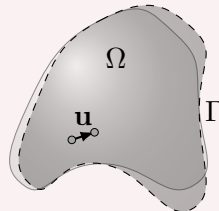
Statement of the linear elastodynamics problem

- **Unknown:** displacement $\mathbf{u}(\mathbf{x}, t) \in \mathbb{R}^3$

Undeformed solid



Deformed solid



- PDE governing the evolution of the displacements $\mathbf{u}$:

$$\nabla^T \mathbf{C} \nabla \mathbf{u}(\mathbf{x}, t) + \mathbf{f}(\mathbf{x}, t) = \rho \ddot{\mathbf{u}}(\mathbf{x}, t) \quad \forall (\mathbf{x}, t) \in \Omega \times]0, T[$$

Strong and weak forms of elastodynamics

Strong form

PDE:
$$\nabla^T \mathbf{C} \nabla \mathbf{u} + \mathbf{f} = \rho \ddot{\mathbf{u}}$$

BC on Γ_u :
$$\mathbf{u} = \hat{\mathbf{u}}$$

BC on Γ_σ :
$$\mathbf{N}^T \mathbf{C} \nabla \mathbf{u} = \hat{\mathbf{f}}$$

IC at $t = 0$:
$$\mathbf{u} = \mathbf{u}_0, \dot{\mathbf{u}} = \mathbf{v}_0$$

Weak form

Virtual work:

$$\int_{\Omega} (\nabla \delta \mathbf{u})^T \mathbf{C} \nabla \mathbf{u} \, d\Omega + \int_{\Omega} \rho \delta \mathbf{u}^T \ddot{\mathbf{u}} \, d\Omega = \int_{\Gamma_\sigma} \delta \mathbf{u}^T \hat{\mathbf{f}} \, d\Gamma + \int_{\Omega} \delta \mathbf{u}^T \mathbf{f} \, d\Omega$$

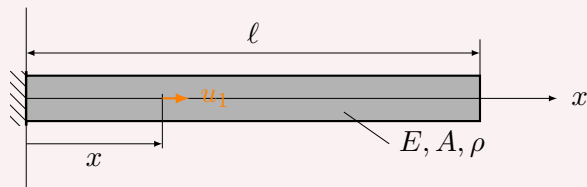
Functional spaces:

$$\begin{aligned} \mathcal{U} &= \{ \mathbf{u}(t, \cdot) \in H^1(\Omega) \mid \mathbf{u} = \hat{\mathbf{u}} \text{ on } \Gamma_u \} \\ \mathcal{V} &= \{ \delta \mathbf{u} \in H^1(\Omega) \mid \delta \mathbf{u} = \mathbf{0} \text{ on } \Gamma_u \} \end{aligned}$$

IC at $t = 0$:

$$\mathbf{u} = \mathbf{u}_0, \dot{\mathbf{u}} = \mathbf{v}_0$$

Strong form for longitudinal vibrations of a bar



- A cross-sectional area
- ℓ length
- E Young's modulus (isotropic bar)
- ρ material density
- u_1 axial displacement
- x axial coordinate

Find $u_1 \in C^2([0, \ell] \times [0, T])$ such that

$$EA \partial_{xx}^2 u_1(x, t) = \rho A \ddot{u}_1(x, t)$$

boundary conditions:

$$u_1(0, t) = 0$$

$$EA \partial_x u_1(\ell, t) = 0$$

initial conditions:

$$u_1(x, 0) = u_0(x)$$

$$\dot{u}_1(x, 0) = v_0(x)$$

Weak form for longitudinal vibrations of a bar

Find the displacement $u_1 \in \mathcal{U}$ such that for any virtual displacement $\delta u_1 \in \mathcal{V}$ we have

$$\int_0^\ell EA \partial_x u_1 \partial_x (\delta u_1) dx + \int_0^\ell \rho A \ddot{u}_1 \delta u_1 dx = 0,$$

where

$$\mathcal{U} = \{u_1(\cdot, t) \in H^1(]0, \ell[) \mid u_1(0, t) = 0 \ \forall t \in]0, T[\}$$

$$\mathcal{V} = \{\delta u_1 \in H^1(]0, \ell[) \mid \delta u_1(0) = 0\}$$

Moreover, the initial conditions read

$$u_1(x, 0) = u_0(x),$$

$$\dot{u}_1(x, 0) = v_0(x).$$

Further evidence in favor of the weak form

Weighted residuals

- Let $\mathbf{u}$ be a solution of

$$\nabla^T \mathbf{C} \nabla \mathbf{u} + \mathbf{f} - \rho \ddot{\mathbf{u}} = 0.$$

- Let $\mathbf{u}^h$ an approximate solution.

In general $\mathbf{u}^h$ does not satisfy the differential equation and hence results in an error or a *residual*:

$$\nabla^T \mathbf{C} \nabla \mathbf{u}^h + \mathbf{f} - \rho \ddot{\mathbf{u}}^h = \mathbf{R}^h.$$

- We impose that the residual is zero in a certain Euclidean vector space $\mathcal{V}^h$ of finite dimension. Thus

$$\langle \mathbf{R}^h, \mathbf{v}^h \rangle = 0 \quad \text{for all } \mathbf{v}^h \in \mathcal{V}^h$$

where $\langle \cdot, \cdot \rangle$ denotes the scalar product.

Weighted residual

- For vector-valued functions, there is a natural definition of scalar product:

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_{\Omega} \mathbf{f}^T \mathbf{g} \, d\Omega.$$

- Hence $\langle \mathbf{R}^h, \mathbf{v}^h \rangle = 0$ for all $\mathbf{v}^h \in \mathcal{V}^h$ implies

$$\int_{\Omega} (\mathbf{v}^h)^T (\nabla^T \mathbf{C} \nabla \mathbf{u}^h + \mathbf{f} - \rho \ddot{\mathbf{u}}^h) \, d\Omega = 0 \quad \text{for all } \mathbf{v}^h \in \mathcal{V}^h.$$

- Applying the divergence theorem results into the equation of the weak form:

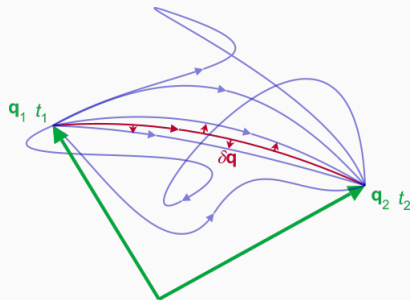
$$\int_{\Omega} (\nabla \mathbf{v}^h)^T \mathbf{C} \nabla \mathbf{u}^h \, d\Omega + \int_{\Omega} \rho (\mathbf{v}^h)^T \ddot{\mathbf{u}}^h \, d\Omega = \int_{\Gamma_{\sigma}} (\mathbf{v}^h)^T \hat{\mathbf{f}} \, d\Gamma + \int_{\Omega} (\mathbf{v}^h)^T \mathbf{f} \, d\Omega \quad \forall \mathbf{v}^h \in \mathcal{V}^h.$$

Hamilton's principle - principle of stationary action

The path taken by the system is defined by the admissible function $\mathbf{u}$ that makes the functional stationary:

$$J = \int_{t_1}^{t_2} T(\dot{\mathbf{u}}) - U(\mathbf{u}) + W(\mathbf{u}) dt.$$

- T is the total kinetic energy,
- U represents the potential (elastic) energy of the flexible structure,
- W the work done by external loads that are acting on the body.



Credit: Wikipedia - Hamilton's principle

Hamilton's principle - principle of stationary action.

- The kinetic energy associated with a flexible body that has volume Ω is given as

$$T(\dot{\mathbf{u}}) = \frac{1}{2} \int_{\Omega} \rho \dot{\mathbf{u}}^T \dot{\mathbf{u}} d\Omega$$

- The total elastic energy of a deformable structure is given as

$$U(\mathbf{u}) = \frac{1}{2} \int_{\Omega} \boldsymbol{\varepsilon}^T \mathbf{C} \boldsymbol{\varepsilon} d\Omega = \frac{1}{2} \int_{\Omega} (\nabla \mathbf{u})^T \mathbf{C} \nabla \mathbf{u} d\Omega$$

where $\boldsymbol{\varepsilon}$ is the elastic strain and $\mathbf{C}$ is the stiffness material matrix.

- The total work W done by the external mechanical loading is given by

$$W(\mathbf{u}) = \int_{\Omega} \mathbf{u}^T \mathbf{f} d\Omega + \int_{\Gamma_{\sigma}} \mathbf{u}^T \hat{\mathbf{f}} d\Gamma$$

Hamilton's principle

- Computing the first variation δJ , and imposing $\delta J = 0$, yields to:

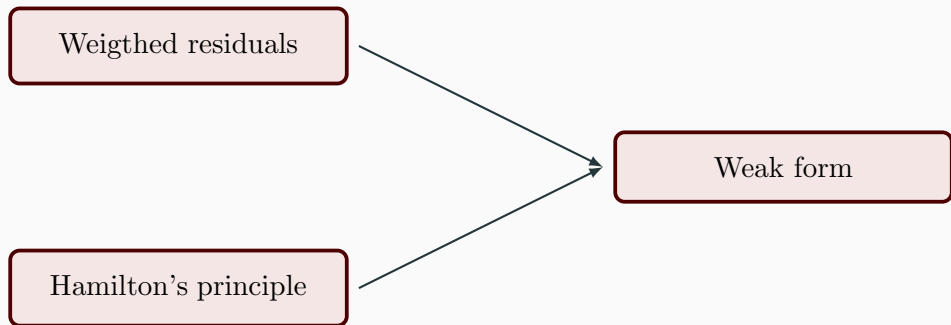
$$\int_{t_1}^{t_2} \left[\int_{\Omega} -\rho(\delta \mathbf{u})^T \ddot{\mathbf{u}} d\Omega - (\delta \boldsymbol{\varepsilon})^T \mathbf{C} \boldsymbol{\varepsilon} + (\delta \mathbf{u})^T \mathbf{f} d\Omega + \int_{\Gamma_{\sigma}} (\delta \mathbf{u})^T \hat{\mathbf{f}} d\Gamma \right] dt = 0$$

for every virtual displacement $\delta \mathbf{u}$ which satisfies $\delta \mathbf{u} = 0$ on Γ_u .

- This implies the weak form of elastodynamics:

$$\int_{\Omega} (\nabla \delta \mathbf{u})^T \mathbf{C} \nabla \mathbf{u} d\Omega + \int_{\Omega} \rho(\delta \mathbf{u})^T \ddot{\mathbf{u}} d\Omega = \int_{\Gamma_{\sigma}} (\delta \mathbf{u})^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} (\delta \mathbf{u})^T \mathbf{f} d\Omega \quad \forall \delta \mathbf{u}.$$

Weak form: a cornerstone concept



Galerkin method

General ideas of Galerkin methods



- Galerkin methods are a class of numerical techniques used to transform differential equations in weak formulation, into discrete problems.
- The approximate solution is determined via a finite set of **basis functions**.
- Galerkin approximation compute the best possible approximate solution among a family of potential solutions.



Boris Galerkin 1871 - 1945

Approximate solution of the weak form

Finite-dimensional functional spaces

Choose subspaces $\mathcal{U}^h \subset \mathcal{U}$ and $\mathcal{V}^h \subset \mathcal{V}$ of **dimension n** and solve the *projected weak form* into such subspaces. Moreover, we impose $\mathcal{U}^h = \mathcal{V}^h$.

Displacement approximation

Instead of searching for $\mathbf{u} \in \mathcal{U}$ such that the weak form is satisfied for any $\delta \mathbf{u} \in \mathcal{V}$, we shall search for $\mathbf{u}^h \in \mathcal{U}^h$ such that the weak form is satisfied for any $\delta \mathbf{u}^h \in \mathcal{V}^h$.

$$\mathbf{u}(\mathbf{x}, t) \approx \mathbf{u}^h(\mathbf{x}, t) \in \mathcal{U}^h \subset \mathcal{U}$$

$$\delta \mathbf{u}(\mathbf{x}) \approx \delta \mathbf{u}^h(\mathbf{x}) \in \mathcal{V}^h \subset \mathcal{V}$$

The key property of the Galerkin approach is that the error is *orthogonal* to the chosen subspaces.

Shape functions

Let $\mathbf{u}^h(\mathbf{x}, t) = \mathbf{H}(\mathbf{x})\mathbf{q}(t)$ and $\delta\mathbf{u}^h(\mathbf{x}) = \mathbf{H}(\mathbf{x})\delta\mathbf{q}$ where

- $\mathbf{H}(\mathbf{x})$ is a $3 \times n$ matrix of **shape functions**, defined *globally* on Ω .
- $\mathbf{q}(t)$ is a $n \times 1$ vector of (*unknown*) time-dependent functions.
- $\delta\mathbf{q}$ is a $n \times 1$ vector of constants.

Shape functions are linearly independent: they form a basis of $\mathcal{U}^h = \mathcal{V}^h$.

$$\begin{pmatrix} u_1^h(\mathbf{x}, t) \\ u_2^h(\mathbf{x}, t) \\ u_3^h(\mathbf{x}, t) \end{pmatrix} = \begin{bmatrix} h_{11}(\mathbf{x}) & h_{12}(\mathbf{x}) & \dots & h_{1n}(\mathbf{x}) \\ h_{21}(\mathbf{x}) & h_{22}(\mathbf{x}) & \dots & h_{2n}(\mathbf{x}) \\ h_{31}(\mathbf{x}) & h_{32}(\mathbf{x}) & \dots & h_{3n}(\mathbf{x}) \end{bmatrix} \begin{pmatrix} q_1(t) \\ q_2(t) \\ \vdots \\ q_n(t) \end{pmatrix}$$

$$\begin{pmatrix} \delta u_1^h(\mathbf{x}) \\ \delta u_2^h(\mathbf{x}) \\ \delta u_3^h(\mathbf{x}) \end{pmatrix} = \begin{bmatrix} h_{11}(\mathbf{x}) & h_{12}(\mathbf{x}) & \dots & h_{1n}(\mathbf{x}) \\ h_{21}(\mathbf{x}) & h_{22}(\mathbf{x}) & \dots & h_{2n}(\mathbf{x}) \\ h_{31}(\mathbf{x}) & h_{32}(\mathbf{x}) & \dots & h_{3n}(\mathbf{x}) \end{bmatrix} \begin{pmatrix} \delta q_1 \\ \delta q_2 \\ \vdots \\ \delta q_n \end{pmatrix}$$

Approximate solution of the weak form

- Find $\mathbf{u}^h \in \mathcal{U}^h$ such that for all $\delta \mathbf{u}^h \in \mathcal{V}^h$ we have

$$\int_{\Omega} (\nabla \delta \mathbf{u}^h)^T \mathbf{C} \nabla \mathbf{u}^h d\Omega + \int_{\Omega} \rho (\delta \mathbf{u}^h)^T \ddot{\mathbf{u}}^h d\Omega = \int_{\Gamma_{\sigma}} (\delta \mathbf{u}^h)^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} (\delta \mathbf{u}^h)^T \mathbf{f} d\Omega$$

- Find $\mathbf{q}(t) \in \mathbb{R}^n$ such that for all $\delta \mathbf{q} \in \mathbb{R}^n$ we have

$$\int_{\Omega} (\nabla \mathbf{H} \delta \mathbf{q})^T \mathbf{C} \nabla \mathbf{H} \mathbf{q}(t) d\Omega + \int_{\Omega} \rho (\mathbf{H} \delta \mathbf{q})^T \mathbf{H} \ddot{\mathbf{q}}(t) d\Omega = \int_{\Gamma_{\sigma}} (\mathbf{H} \delta \mathbf{q})^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} (\mathbf{H} \delta \mathbf{q})^T \mathbf{f} d\Omega$$

- Rearranging the terms we obtain the following expression:

$$\delta \mathbf{q}^T \left[\underbrace{\left(\int_{\Omega} (\nabla \mathbf{H})^T \mathbf{C} \nabla \mathbf{H} d\Omega \right)}_{\mathbf{K}} \mathbf{q}(t) + \underbrace{\left(\int_{\Omega} \rho \mathbf{H}^T \mathbf{H} d\Omega \right)}_{\mathbf{M}} \ddot{\mathbf{q}}(t) - \underbrace{\left(\int_{\Gamma_{\sigma}} \mathbf{H}^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} \mathbf{H}^T \mathbf{f} d\Omega \right)}_{\mathbf{r}(t)} \right] = 0$$

- **Stiffness matrix** ($n \times n$):

$$\mathbf{K} = \int_{\Omega} \mathbf{B}^T \mathbf{C} \mathbf{B} d\Omega$$

where $\mathbf{B}$ is the $(6 \times n)$ deformation matrix defined by $\mathbf{B} = \nabla \mathbf{H}$.

- **Mass matrix** ($n \times n$):

$$\mathbf{M} = \int_{\Omega} \rho \mathbf{H}^T \mathbf{H} d\Omega.$$

- **Applied forces vector** ($n \times 1$):

$$\mathbf{r}(t) = \int_{\Gamma_{\sigma}} \mathbf{H}^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} \mathbf{H}^T \mathbf{f} d\Omega.$$

Semi-discrete weak form of elastodynamics

Given Ω , Γ , $\mathbf{C}$, ρ , $\mathbf{f}$, $\hat{\mathbf{u}}$, $\hat{\mathbf{f}}$, $\mathbf{u}_0$, $\mathbf{v}_0$, and a matrix of shape functions $\mathbf{H}$, find the vector $\mathbf{q} \in C^2([0, T], \mathbb{R}^n)$ such that for every vector $\delta \mathbf{q} \in \mathbb{R}^n$ we have

$$\delta \mathbf{q}^T [\mathbf{M} \ddot{\mathbf{q}}(t) + \mathbf{K} \mathbf{q}(t) - \mathbf{r}(t)] = 0$$

coupled with initial conditions: $\mathbf{q}(0) = \mathbf{q}_0$ and $\dot{\mathbf{q}}(0) = \mathbf{p}_0$.

Remark. This is equivalent to solving the system of second-order ODEs:

$$\mathbf{M} \ddot{\mathbf{q}}(t) + \mathbf{K} \mathbf{q}(t) = \mathbf{r}(t)$$

with initial conditions $\mathbf{q}(0) = \mathbf{q}_0$ and $\dot{\mathbf{q}}(0) = \mathbf{p}_0$.

Treatment of boundary conditions

- Boundary conditions on Γ_u :

$$\mathbf{u}^h = \hat{\mathbf{u}} \quad \text{and} \quad \delta \mathbf{u}^h = \mathbf{0}$$

Consequently, when defining shape functions, it is *necessary to impose* that

$$\mathbf{H}(\mathbf{x})\mathbf{q}(t) = \hat{\mathbf{u}}(\mathbf{x}, t) \quad \text{and} \quad \mathbf{H}(\mathbf{x})\delta\mathbf{q} = \mathbf{0} \quad \text{on } \Gamma_u \text{ and for any } t \in]0, T[.$$

Treatment of boundary conditions

- Boundary conditions on Γ_u :

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Consequently, when defining shape functions, it is *necessary to impose* that

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- Boundary condition on Γ_σ :

$$\mathbf{N}^T \mathbf{C} \nabla \mathbf{u}^h = \hat{\mathbf{f}}$$

Does not affect the choice of shape functions since this condition has already been used in the derivation of the weak form.

Treatment of initial conditions (at $t = 0$)

Recall that $\mathbf{u}(t = 0) = \mathbf{u}_0$ and $\dot{\mathbf{u}}(t = 0) = \mathbf{v}_0$.

$$\int_{\Omega} \rho(\delta \mathbf{u}^h)^T \mathbf{u}|_{t=0} d\Omega = \int_{\Omega} \rho(\delta \mathbf{u}^h)^T \mathbf{u}_0 d\Omega$$

$$\int_{\Omega} \rho(\delta \mathbf{u}^h)^T \dot{\mathbf{u}}|_{t=0} d\Omega = \int_{\Omega} \rho(\delta \mathbf{u}^h)^T \mathbf{v}_0 d\Omega$$

Substituting $\mathbf{u}^h(\mathbf{x}, t) = \mathbf{H}(x)\mathbf{q}(t)$ and $\delta \mathbf{u}^h(x) = \mathbf{H}(x)\delta \mathbf{q}$ gives

$$\delta \mathbf{q}^T \left(\underbrace{\int_{\Omega} \rho \mathbf{H}^T \mathbf{H} d\Omega}_{\mathbf{M}} \right) \mathbf{q}(0) = \delta \mathbf{q}^T \left(\int_{\Omega} \rho \mathbf{H}^T \mathbf{u}_0 d\Omega \right)$$

$$\delta \mathbf{q}^T \left(\underbrace{\int_{\Omega} \rho \mathbf{H}^T \mathbf{H} d\Omega}_{\mathbf{M}} \right) \dot{\mathbf{q}}(0) = \delta \mathbf{q}^T \left(\int_{\Omega} \rho \mathbf{H}^T \mathbf{v}_0 d\Omega \right)$$

$$\mathbf{q}(0) = \underbrace{\mathbf{M}^{-1} \left(\int_{\Omega} \rho \mathbf{H}^T \mathbf{u}_0 d\Omega \right)}_{\mathbf{q}_0}$$

$$\dot{\mathbf{q}}(0) = \underbrace{\mathbf{M}^{-1} \left(\int_{\Omega} \rho \mathbf{H}^T \mathbf{v}_0 d\Omega \right)}_{\mathbf{p}_0}$$

Static, modal, and transient analysis

- ① **Static analysis:** determines deformations $\mathbf{q}$ due to constant loads.

$$\mathbf{K}\mathbf{q} = \mathbf{r}$$

- ② **Modal analysis:** studies dynamic properties in the frequency domain.

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{0}$$

Natural frequency: key structural property essential in structural design.

- Lower frequencies $\rightarrow$ higher displacement amplitudes $\rightarrow$ more dangerous.
- Resonance when external excitation frequency matches a natural frequency.
- Prolonged resonance $\rightarrow$ catastrophic failure.

- ③ **Transient analysis:** examines time-dependent structural responses $\mathbf{q}(t)$ to time-dependent excitations $\mathbf{r}(t)$.

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{r}(t)$$

Modal analysis

- Free vibrations: $\mathbf{f} = 0$, thus $\mathbf{r}(t) = \mathbf{0}$.
- Assume harmonic solution:

$$\mathbf{q}(t) = \mathbf{p}\alpha \cos(\omega t - \varphi)$$

- $\mathbf{p}$ is the mode shape (eigenvector).
 - ω is the natural angular frequency (eigenvalue) [rad/s].
 - Regular frequency $f = \omega/2\pi$ [Hz].
 - α (amplitude) and φ (phase) depend on initial conditions.
- Leads to the eigenvalue problem:

$$(\mathbf{K} - \omega^2\mathbf{M})\mathbf{p} = \mathbf{0}.$$

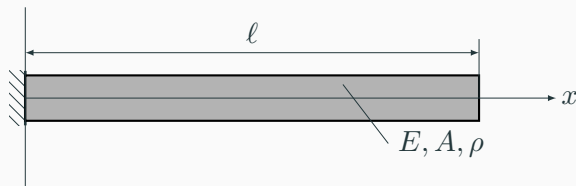
- Non-trivial solutions exist only if:

$$\det(\mathbf{K} - \omega^2\mathbf{M}) = 0.$$

The eigenvalues $\lambda_i = \omega_i^2$ are the squares of the natural angular frequencies. The eigenvectors $\mathbf{p}_i$ are the corresponding mode shapes.

Example: longitudinal vibration of a bar

Example - Galerkin approximation of longitudinal vibrations of a bar



- A cross-sectional area
- ℓ length
- E Young's modulus
- ρ material density
- u_1 axial displacement
- x axial coordinate

Objective: find a Galerkin approximation of the axial displacement u_1 using appropriate functional spaces.

Galerkin approximation of longitudinal vibrations of a bar

Substituting

$$u_1^h(x, t) = \mathbf{H}(x)\mathbf{q}(t) = \sum_{i=1}^n h_i(x)q_i(t) \quad \text{and} \quad \delta u_1^h(x) = \mathbf{H}(x)\delta\mathbf{q} = \sum_{i=1}^n h_i(x)\delta q_i$$

into the integral equation

$$\int_0^\ell EA \partial_x u_1^h \partial_x (\delta u_1^h) dx + \int_0^\ell \rho A \ddot{u}_1^h \delta u_1^h dx = 0,$$

allow us to write

$$\mathbf{K}\mathbf{q}(t) + \mathbf{M}\ddot{\mathbf{q}}(t) = 0$$

where

$$k_{ij} = \int_0^\ell EA h_i'(x) h_j'(x) dx, \quad m_{ij} = \int_0^\ell \rho A h_i(x) h_j(x) dx.$$

- Boundary conditions:

$$u_1^h(0, t) = 0 \quad \text{and} \quad \delta u_1^h(0, t) = 0$$

Therefore $\mathbf{H}(0) = \mathbf{0}$.

Boundary and initial conditions

■ Boundary conditions:

$$u_1^h(0, t) = 0 \quad \text{and} \quad \delta u_1^h(0, t) = 0$$

Therefore $\mathbf{H}(0) = \mathbf{0}$.

■ Initial conditions:

$$u_1^h(x, 0) = u_0(x) \quad \text{and} \quad \dot{u}_1^h(x, 0) = v_0(x)$$

Thus $\mathbf{q}_1(0) = \mathbf{q}_0 = \mathbf{M}^{-1} \int_{\Omega} \rho \mathbf{H}^T u_0 \, d\Omega$ and $\dot{\mathbf{q}}_1(0) = \mathbf{p}_0 = \mathbf{M}^{-1} \int_{\Omega} \rho \mathbf{H}^T v_0 \, d\Omega$.

One-term Galerkin approximation

- Let $n = 1$ then

$$u_1^h(x, t) = \mathbf{H}(x)\mathbf{q}(t) = h_1(x)q_1(t)$$

$$\delta u_1^h(x) = \mathbf{H}(x)\delta\mathbf{q} = h_1(x)\delta q_1$$

where we choose

$$h_1(x) = \frac{x}{\ell}.$$

Notice that $h_1 \in H^1(]0, \ell[)$ and $h_1(0) = 0$.

- The semi-discrete weak form is: find the function $q_1(t)$ such that:

$$k_{11}q_1(t) + m_{11}\ddot{q}_1(t) = 0$$

$$k_{11} = \int_0^\ell EA \left(\frac{1}{\ell}\right)^2 dx = \frac{EA}{\ell}, \quad m_{11} = \int_0^\ell \rho A \left(\frac{x}{\ell}\right)^2 dx = \frac{\rho A \ell}{3}$$

Second order differential equation

The linear homogeneous second order ODE with constant coefficients governing the free vibration of single degree of freedom conservative system:

$$\begin{cases} k_{11}q_1(t) + m_{11}\ddot{q}_1(t) = 0 \\ q_1(0) = q_0 \\ \dot{q}_1(0) = p_0 \end{cases}$$

admits an unique solution given by

$$q_1(t) = p \cos(\omega_1 t - \varphi),$$

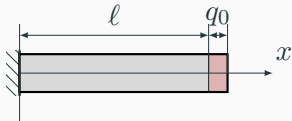
where the approx fundamental frequency is

$$\omega_1 = \sqrt{k_{11}/m_{11}} = \sqrt{3}\sqrt{E/\rho\ell^2},$$

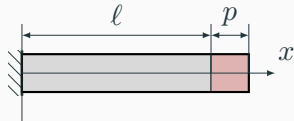
moreover $p = \sqrt{q_0^2 + (p_0/\omega_1)^2}$ and $\tan(\varphi) = p_0/q_0\omega_1$.

Displacement approximation

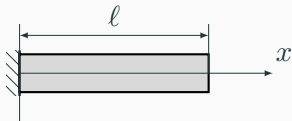
$$u_1(x, t) = h_1(x)q_1(t) = \frac{px}{\ell} \cos(\omega_1 t - \varphi)$$



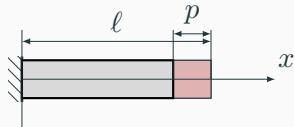
(a) $t = 0$



(b) $t = (\varphi + 2k\pi)/\omega_1$



(c) $t = (\varphi + \pi/2 + k\pi)/\omega_1$



(d) $t = (\varphi + (2k + 1)\pi)/\omega_1$

Fundamental frequency comparison

- Approximated fundamental frequency: (obtained via one-term Galerkin)

$$\omega_1 = \sqrt{\frac{k_{11}}{m_{11}}} = \sqrt{3} \sqrt{\frac{E}{\rho \ell^2}}$$

- Exact fundamental frequency:

$$\omega_1^e = \frac{\pi}{2} \sqrt{\frac{E}{\rho \ell^2}}$$

- Relative error:

$$\text{relative error} = \frac{|\omega_1 - \omega_1^e|}{\omega_1^e} = 10.3\%$$

One-term quadratic Galerkin approximation

Let $n = 1$ and

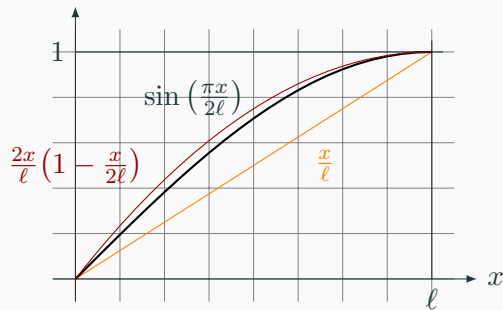
$$u_1^h(x, t) = \mathbf{H}(x)\mathbf{q}(t) = h_1(x)q_1(t)$$

$$\delta u_1^h(x) = \mathbf{H}(x)\delta\mathbf{q} = h_1(x)\delta q_1$$

where the shape function is chosen as

$$h_1(x) = \frac{2x}{\ell} \left(1 - \frac{x}{2\ell}\right).$$

Notice $h_1 \in H^1(]0, \ell[)$ and $h_1(0) = 0$.



One-term quadratic Galerkin approximation

- The stiffness and mass '*matrices*' are

$$k_{11} = \int_0^\ell \frac{4EA}{\ell^2} \left(1 - \frac{x}{\ell}\right)^2 dx = \frac{4EA}{3\ell}$$
$$m_{11} = \int_0^\ell \frac{4\rho Ax^2}{\ell^2} \left(1 - \frac{x}{2\ell}\right)^2 dx = \frac{8\rho A\ell}{15}$$

- Approximated fundamental frequency: (obtained via one-term quadratic Galerkin)

$$\omega_1 = \sqrt{\frac{k_{11}}{m_{11}}} = \sqrt{\frac{5}{2}} \sqrt{\frac{E}{\rho\ell^2}}$$

- Relative error:

$$\text{relative error} = \frac{|\omega_1 - \omega_1^e|}{\omega_1^e} = 0.7\%$$

n -terms Galerkin approximation

► [Go to Matlab Drive](#)

Advantages and drawbacks of Galerkin method

In a nutshell: Galerkin method transforms the partial differential equations (PDE), expressed in their weak formulation, into a system of ordinary differential equations (ODE).

Advantages:

- ✓ Converges quickly with appropriate shape functions.
- ✓ Provides a systematic and structured approach for approximating solutions.
- ✓ The same set of functions is used to express real and virtual variables.

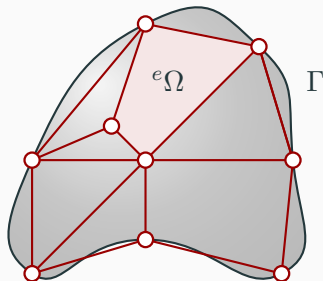
Drawbacks:

- ✗ Accuracy heavily dependent on choice of basis functions.
- ✗ No physical interpretation of the unknown variable $\mathbf{q}(t)$.
- ✗ The formulation of initial and boundary conditions in the discretized form is cumbersome.

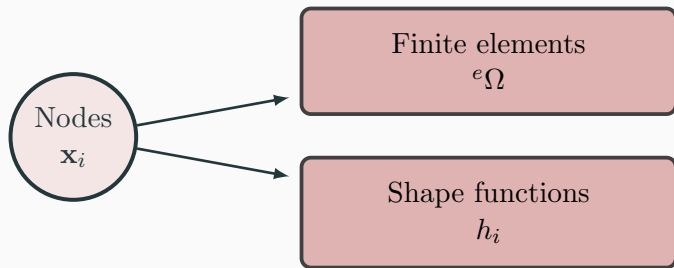
Finite element method in global coordinate system

Divide and conquer mentality

- The geometry of a structure is discretised when it is split into a mesh of finite elements $^1\Omega, \dots, ^m\Omega$ (smaller components or regular shapes).
- The discretisation introduces an approximation error which can be reduced by using a finer mesh (i.e. more elements), or by increasing the accuracy of the finite elements chosen.
- Discretisation starts with the definition of a **set of nodes**: $\mathbf{x}_i, i = 1, \dots, p$.



Nodes, elements and shape functions



- **Finite element method:** a special case of the Galerkin method, where shape functions are systematically defined in terms of nodes.
- Each shape function is non-zero only on a very limited number of finite elements (*compact support*).
- The computation of the mass and stiffness matrix is enhanced by the local nature of the shape functions.

Displacements approximation

Let p the number of nodes of the mesh.

$$\mathbf{u}^h(\mathbf{x}, t) = \mathbf{H}(\mathbf{x})\mathbf{q}(t)$$

$$\delta\mathbf{u}^h(\mathbf{x}) = \mathbf{H}(\mathbf{x})\delta\mathbf{q}$$

- $\mathbf{H}(\mathbf{x})$ is a $3 \times 3p$ matrix of **shape functions**.

$$\mathbf{H} = \left[\begin{array}{c|c|c|c|c|c} h_1\mathbf{I} & h_2\mathbf{I} & \dots & h_i\mathbf{I} & \dots & h_p\mathbf{I} \end{array} \right]$$

$\mathbf{I}$ is the 3×3 identity matrix.

- $\mathbf{q}(t)$ is a $3p \times 1$ vector of (*unknown*) time-dependent functions.
- $\delta\mathbf{q}$ is a $3p \times 1$ vector of constants.

Displacements approximation - matrix notation

$$\begin{pmatrix} u_1^h \\ u_2^h \\ u_3^h \end{pmatrix} = \begin{bmatrix} h_1 & 0 & 0 & | & h_2 & 0 & 0 & | & \dots & | & h_p & 0 & 0 \\ 0 & h_1 & 0 & | & 0 & h_2 & 0 & | & \dots & | & 0 & h_p & 0 \\ 0 & 0 & h_1 & | & 0 & 0 & h_2 & | & \dots & | & 0 & 0 & h_p \end{bmatrix} \begin{pmatrix} q_{1,1} \\ q_{1,2} \\ q_{1,3} \\ \vdots \\ q_{p,1} \\ q_{p,2} \\ q_{p,3} \end{pmatrix}$$

$$\begin{pmatrix} \delta u_1^h \\ \delta u_2^h \\ \delta u_3^h \end{pmatrix} = \begin{bmatrix} h_1 & 0 & 0 & | & h_2 & 0 & 0 & | & \dots & | & h_p & 0 & 0 \\ 0 & h_1 & 0 & | & 0 & h_2 & 0 & | & \dots & | & 0 & h_p & 0 \\ 0 & 0 & h_1 & | & 0 & 0 & h_2 & | & \dots & | & 0 & 0 & h_p \end{bmatrix} \begin{pmatrix} \delta q_{1,1} \\ \delta q_{1,2} \\ \delta q_{1,3} \\ \vdots \\ \delta q_{p,1} \\ \delta q_{p,2} \\ \delta q_{p,3} \end{pmatrix}$$

Displacements approximation - index notation

$$\mathbf{u}^h(\mathbf{x}, t) = \sum_{i=1}^p h_i(\mathbf{x}) \mathbf{q}_i(t)$$

$$\delta \mathbf{u}^h(\mathbf{x}, t) = \sum_{i=1}^p h_i(\mathbf{x}) \delta \mathbf{q}_i$$

$$\mathbf{q}_i(t) = \begin{pmatrix} q_{i,1}(t) \\ q_{i,2}(t) \\ q_{i,3}(t) \end{pmatrix} \quad \text{and} \quad \delta \mathbf{q}_i = \begin{pmatrix} \delta q_{i,1} \\ \delta q_{i,2} \\ \delta q_{i,3} \end{pmatrix}$$

Deformation, stiffness, mass matrices and loads vector

- $\mathbf{B} = \nabla \mathbf{H}$ is a $(6 \times 3p)$ matrix:

$$\mathbf{B} = \left[\begin{array}{ccc|ccc} \partial_x h_1 & 0 & 0 & \dots & \partial_x h_p & 0 & 0 \\ 0 & \partial_y h_1 & 0 & \dots & 0 & \partial_y h_p & 0 \\ 0 & 0 & \partial_z h_1 & \dots & 0 & 0 & \partial_z h_p \\ 0 & \partial_z h_1 & \partial_y h_1 & \dots & 0 & \partial_z h_p & \partial_y h_p \\ \partial_z h_1 & 0 & \partial_x h_1 & \dots & \partial_z h_p & 0 & \partial_x h_p \\ \partial_y h_1 & \partial_x h_1 & 0 & \dots & \partial_y h_p & \partial_x h_p & 0 \end{array} \right] = [\nabla h_1 \mid \dots \mid \nabla h_p]$$

- $\mathbf{K}$ and $\mathbf{M}$ are $(3p \times 3p)$ matrices and $\mathbf{r}$ is a $(3p \times 1)$ vector:

$$\mathbf{K} = \int_{\Omega} \mathbf{B}^T \mathbf{C} \mathbf{B} d\Omega, \quad \mathbf{M} = \int_{\Omega} \rho \mathbf{H}^T \mathbf{H} d\Omega, \quad \mathbf{r}(t) = \int_{\Gamma_{\sigma}} \mathbf{H}^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} \mathbf{H}^T \mathbf{f} d\Omega.$$

Global nodal shape functions - construction guidelines

Global nodal shape functions $h_i : \Omega \rightarrow \mathbb{R}$ are characterized by the following properties:

- They form a linearly independent basis of polynomials of a given degree.
- Their values lie in the interval $[0, 1]$.
- They satisfy the Kronecker delta property:

$$h_i(\mathbf{x}_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

- The functions h_i vanish on all finite elements that are not adjacent to the node $\mathbf{x}_i$.

Properties of shape functions - convergence criteria

- Continuity of h_i at nodes and interfaces of finite elements.
- Differentiability of h_i inside each finite element.
- Completeness (rigid body motion and constant deformation):

$$\sum_{i=1}^p h_i(\mathbf{x}) = 1 \quad \text{and} \quad \sum_{i=1}^p \nabla h_i(\mathbf{x}) = \mathbf{0}.$$

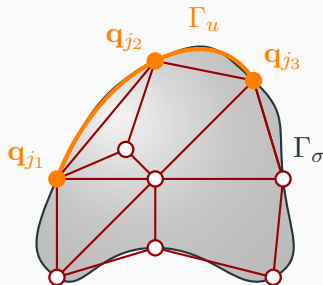
- Approximate real and virtual displacements at the nodes:

$$\mathbf{u}^h(\mathbf{x}_j, t) = \sum_{i=1}^p h_i(\mathbf{x}_j) \mathbf{q}_i(t) = \mathbf{q}_j(t),$$
$$\delta \mathbf{u}^h(\mathbf{x}_j) = \sum_{i=1}^p h_i(\mathbf{x}_j) \delta \mathbf{q}_i = \delta \mathbf{q}_j.$$

Treatment of boundary conditions

On Γ_u we impose $\mathbf{u}^h = \hat{\mathbf{u}}$ and $\delta \mathbf{u}^h = \mathbf{0}$. Consequently for every node $\mathbf{x}_j \in \Gamma_u$:

$$\mathbf{q}_j(t) = \mathbf{u}^h(\mathbf{x}_j, t) = \hat{\mathbf{u}}(\mathbf{x}_j, t) \quad \text{and} \quad \delta \mathbf{q}_j = \delta \mathbf{u}^h(\mathbf{x}_j, t) = \mathbf{0}.$$



- ✓ The j -th component of the approximated nodal displacement $\mathbf{q}$ is known for every node $\mathbf{x}_j \in \Gamma_u$.

Treatment of initial conditions

Since $\mathbf{u}(\mathbf{x}, 0) = \mathbf{u}_0(\mathbf{x})$, enforcing $\mathbf{u}^h(\mathbf{x}, 0) = \mathbf{u}_0(\mathbf{x})$ ensures that the approximate displacement field is consistent with the initial condition. This leads to

$$\mathbf{q}_j(0) = \mathbf{u}^h(\mathbf{x}_j, 0) = \mathbf{u}_0(\mathbf{x}_j).$$

Similarly, imposing the initial velocity condition, $\dot{\mathbf{u}}(\mathbf{x}, 0) = \mathbf{v}_0(\mathbf{x})$, gives

$$\dot{\mathbf{q}}_j(0) = \dot{\mathbf{u}}^h(\mathbf{x}_j, 0) = \mathbf{v}_0(\mathbf{x}_j).$$

- ✓ These expressions provide the initial values of the unknown vector $\mathbf{q}$ directly in terms of the given initial data $\mathbf{u}_0$ and $\mathbf{v}_0$.

Advantages and drawbacks of finite element method

Advantages:

- ✓ Versatile for various engineering problems: mechanics of solids and fluids, dynamics, heat transfer, electrostatics, and more.
- ✓ Provides a systematic algorithm: defines the underlying shape functions used in the approximation.
- ✓ Accuracy control: solution precision can be improved by refining the mesh.

Drawbacks:

- ✗ Choice of mathematical model is crucial: the solution's reliability hinges on selecting an appropriate model.
- ✗ Computational cost: highly refined models increase the computational burden.
- ✗ Requires experience in meshing, boundary conditions, and solver settings.